

Introduction to basic theories of computational tools for studying properties of materials

Presented on 27th July, 2026
in

“5-Day Workshop on AUTOMATED DFT, HIGH- THROUGHPUT & AI-ASSISTED SIMULATIONS ”

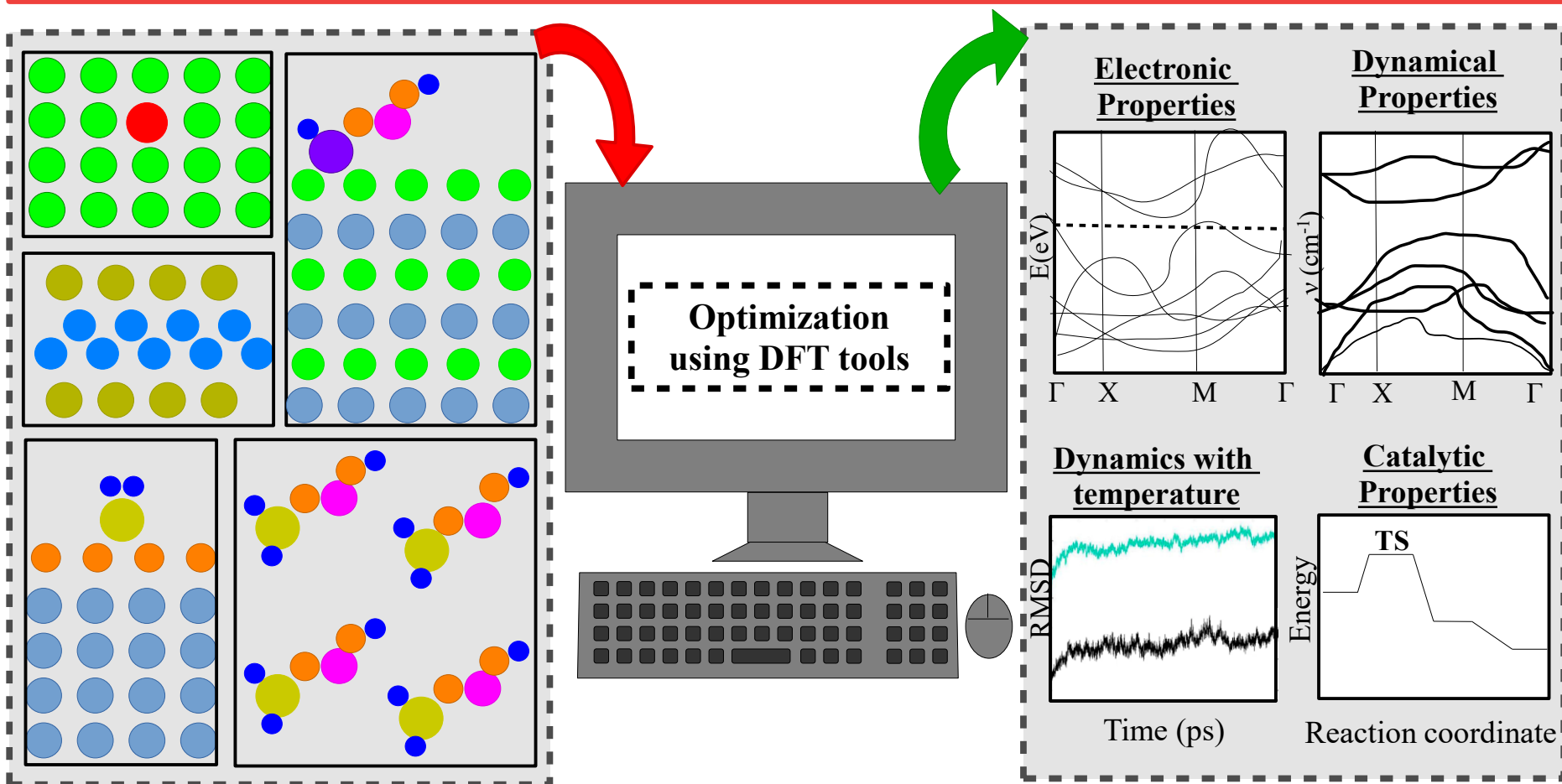
Organized by **MetaChem Academy**, Pune, Maharashtra, India.



and presented by

Niharika Joshi
Asst. Professor,
Fergusson College, Pune,
Maharashtra-411004

SCHEMATIC OF WORKING OF COMPUTATIONAL MODELING AND SIMULATION



Choosing proper optimization tool and set of appropriate analysis will fetch desired results which are feasible for the real-life applications.

Outline

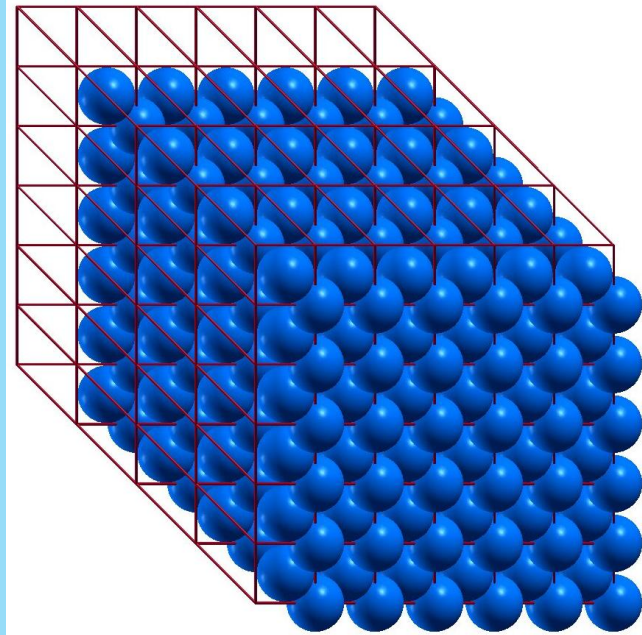
- I. Schrodinger Equation to Density functional theory.
- II. Formulation of DFT: Kohn-Sham Equations
- III. Solving Kohn-Sham Equations: Self-consistent field cycle
- IV. Challenges while solving Kohn-Sham equation and their remedies.
- V. Structure of input file for optimization system in Quantum ESPRESSO.

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- II. Formulation of DFT: Kohn-Sham Equations
- III. Solving Kohn-Sham Equations: Self-consistent field cycle
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- V. Structure of input file for optimization system in Quantum ESPRESSO.

Test system: Diamond

- ✓ Face-centre-cubic lattice with 2 basis atoms.
- ✓ Tetrahedral coordination: Each C covalently binding with 4 C.
- ✓ Lattice parameter = $3.57 \text{ \AA}$
- ✓ C-C = $1.54 \text{ \AA}$



Understanding Density Functional Theory

Building Hamiltonian for a system

Prediction of the electronic and geometric structure of a system requires determination of its quantum-mechanical total energy (E) and subsequent minimization of that energy with respect to the electronic (r_i) and nuclear (R_I) coordinates.

$$\mathbf{H}\Psi(\{R_I, r_i\}) = E\Psi(\{R_I, r_i\})$$

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$$\mathbf{H} = \text{KE of nuclei} + \text{KE of electrons} + \text{e-e Coulomb PE} + \text{Ion-e Coulomb PE} + \text{Ion-ion Coulomb PE}$$

Understanding Density Functional Theory

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$$\begin{aligned} \mathbf{H} = & \text{KE of nuclei} + \text{KE of electrons} + \text{e-e Coulomb PE} \\ & + \text{Ion-e Coulomb PE} + \text{Ion-ion Coulomb PE} \\ \mathbf{H} = & - \sum_I \frac{\hbar^2}{2M_I} \nabla_{R_I}^2 - \sum_i \frac{\hbar^2}{2m_i} \nabla_{r_i}^2 + \sum_{i,j;i>j} \frac{1}{4\pi\epsilon_0} \frac{e^2}{|r_i - r_j|} \\ & - \sum_{i,I} \frac{1}{4\pi\epsilon_0} \frac{Z_I e^2}{|R_I - r_i|} + \sum_{I,J;I>J} \frac{1}{4\pi\epsilon_0} \frac{Z_I Z_J e^2}{|R_I - R_J|} \end{aligned}$$

Understanding Density Functional Theory

Born-Oppenheimer Approximation: Simplifying the problem

- ✓ $m_e \ll M_n$ but F^{ext} on electrons = F^{ext} on nuclei
- ✓ Electrons thus respond much faster than nuclei. Thus, the nuclei can be treated adiabatically.
- ✓ One can separate electronic and nuclear coordinates dependencies in the many-body wave function.
- ✓ This is so-called the Born-Oppenheimer approximation.

Understanding Density Functional Theory

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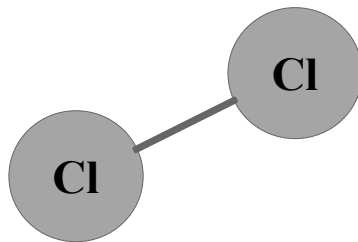
This reduces the many-body problem to the solution of the dynamics of the electrons in some frozen-in configuration of the nuclei.

Electronic Hamiltonian

$$\mathbf{H}_e = - \sum_i \frac{\hbar^2}{2m_i} \nabla_{r_i}^2 + \sum_{i,j;i>j} \frac{1}{4\pi\epsilon_0} \frac{e^2}{|r_i - r_j|} - \sum_{i,I} \frac{1}{4\pi\epsilon_0} \frac{Z_I e^2}{|R_I - r_i|}$$

Understanding Density Functional Theory

Solving the electronic Hamiltonian for real-life systems is also huge task!

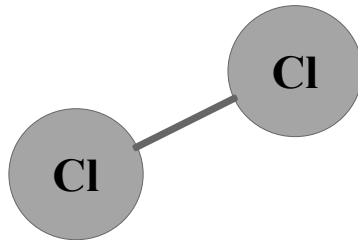


No. of electrons: 34
No. of wavefunctions: $34 * 3 = 102$

- ✓ For determining 102 wavefunctions one must solve 102 equations!
- ✓ e-e interactions are not negligible even at large separation.
- ✓ For determining a single electron wavefunction, rest all other wavefunctions must be known.

Understanding Density Functional Theory

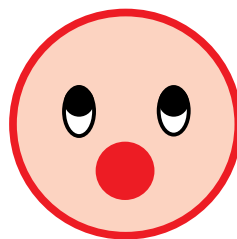
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- ✓ e-e interactions are not negligible even at large separation.
- ✓ For determining a single electron wavefunction, rest all other wavefunctions must be known.
- ✓ Thus, the problem becomes many-body problem.

Now imagine size of problem for a system containing 100 atoms!!!!



Understanding Density Functional Theory

Problem: Working with $\Psi(r)$ is extremely challenging for real-life systems!

Solution: $\rho(r)$

Understanding Density Functional Theory

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Solution: $\rho(r)$

Density Functional Theory

Two postulates proposed by **P. Hohenberg** and **W. Kohn** are the base for DFT.

Postulate 1

For a many-body system the total energy of the system is a unique functional of the electron density.

$$E = E[\rho(r)]$$

Postulate 2:

For a trial electron density, such that $\rho^t(r) \geq 0$ and $\int \rho^t(r) dr = n$, the ground state total energy, $E_0 \leq E[\rho^t(r)]$.

Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

Consider a non-interacting system consisting of n electrons.

$$\mathbf{H}_s = \sum_i^n -\frac{1}{2}\nabla_i^2 + \underbrace{\sum_i^n v_s(\mathbf{r})}_{\text{electron-ion potential}} = \sum_i^n \mathbf{h}_s$$

One-electron
equation:

$$\mathbf{h}_s\psi_i = \left[-\frac{1}{2}\nabla_i^2 + v_s(\mathbf{r})\right]\psi_i = \epsilon_i\psi_i$$

Understanding Density Functional Theory

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Kohn-Sham equations for interacting system:

$$\mathbf{h}\psi_i = \left[-\frac{1}{2}\nabla_i^2 + v_{eff}(\mathbf{r})\right]\psi_i = \epsilon_i\psi_i$$

Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

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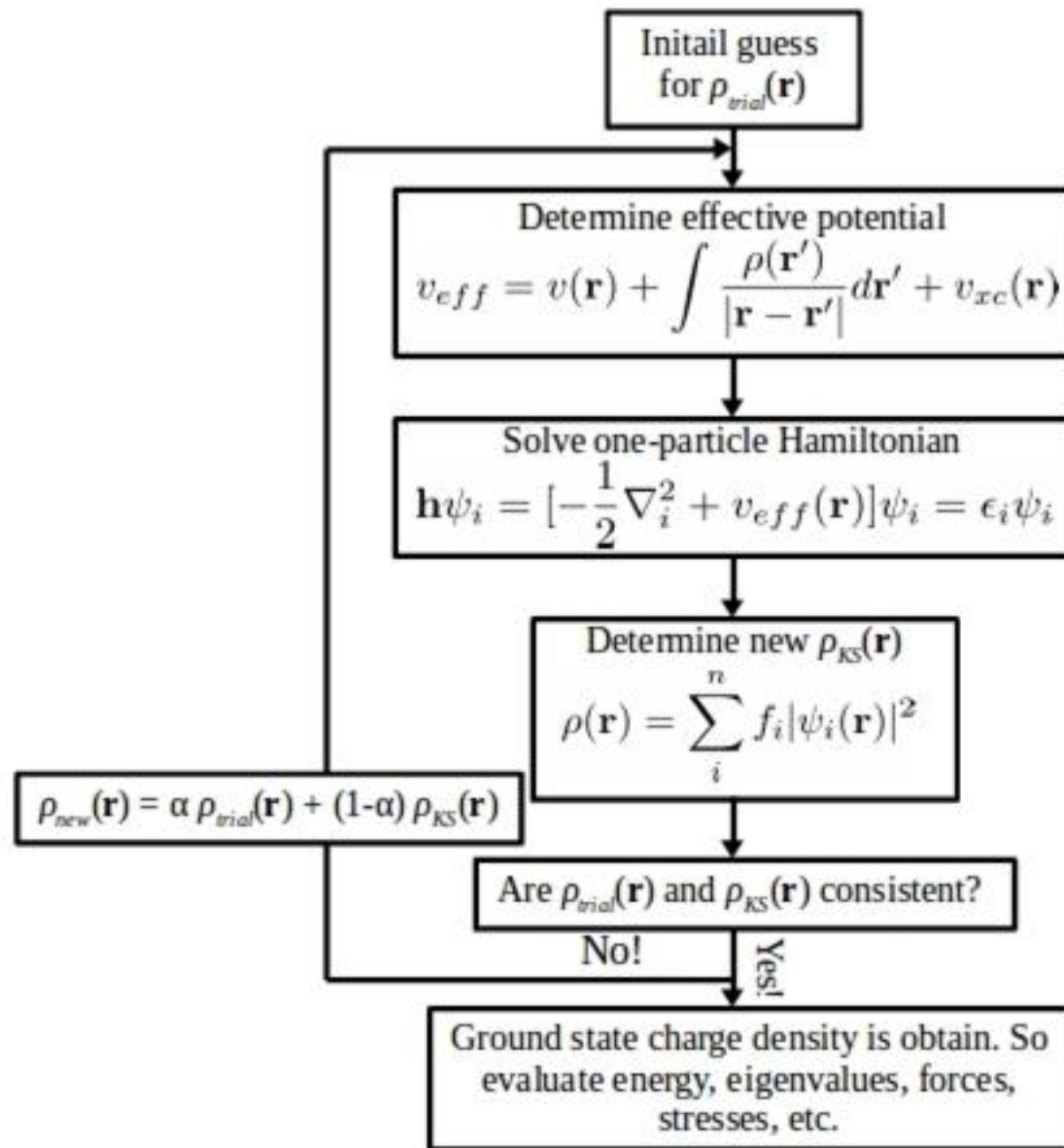
$$v_{eff} = \boxed{v(\mathbf{r})} + \boxed{\int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'} + \boxed{v_{xc}(\mathbf{r})}$$

e-ion potential

e-e Coulomb
potential

e-e exchange-
correlation potential

Self-Consistent Field: Determining ground-state charge density



Structure of QE i/p file

```

&CONTROL
    description of
    the calculation
/
&SYSTEM
    defining structural and
    electronic parameters
/
&ELECTRONS
    parameters for
    electronic cycle
/
&IONS
    parameters for
    ionic cycle
/
&CELL
    parameters for
    cell optimization
/
ATOMIC SPECIES
    name, atomic wt., PP file
ATOMIC_POSITIONS
    atomic coordinates
K_POINTS
    k-point mesh
CELL_PARAMETERS
    cell vectors
    
```

Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

Kohn-Sham equations

$$\mathbf{h}\psi_i = \left[-\frac{1}{2}\nabla_i^2 + v_{eff}(\mathbf{r})\right]\psi_i = \epsilon_i\psi_i$$

$$v_{eff} = \boxed{v(\mathbf{r})} + \boxed{\int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'} + \boxed{v_{xc}(\mathbf{r})}$$

e-ion potential e-e Coulomb potential e-e exchange-correlation potential

Challenges while solving Kohn-Sham equations:

- How to determine ψ_i for infinite electrons?
- How to proceed with infinite basis set?
- How to determine exchange-correlation potential?
- How to determine electron-ion potential in the region close to core region?

Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

1 Calculating ψ_i for infinite electrons.

Bloch theorem: “When the potential in the single particle Hamiltonian has the translational periodicity of the Bravais lattice such that, $V(\mathbf{r} + \mathbf{P}) = V(\mathbf{r})$, the single particle wavefunction have same symmetry up to a phase factor:

$$\psi_{\mathbf{k}}(\mathbf{r} + \mathbf{P}) = e^{i\mathbf{k} \cdot \mathbf{P}} \psi_{\mathbf{k}}(\mathbf{r})$$

OR

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{P}} u(\mathbf{r})$$

where, $\mathbf{P} = p_1 \cdot \mathbf{a}_1 + p_2 \cdot \mathbf{a}_2 + p_3 \cdot \mathbf{a}_3$ is translation vector
 $u(\mathbf{r})$: function with periodicity of Bravais Lattice

Energy terms for only those electrons and ions must be calculated which lie within the primitive unit cell!!!!

Supplementary Discussion

Real Lattice and Reciprocal Lattice

Real lattice and reciprocal lattice are Fourier transforms (inverse) of each other.

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij}$$

$\mathbf{b}_i$: Reciprocal lattice primitive vectors

$\mathbf{a}_j$: Real lattice primitive vectors

δ_{ij} : Kronecker delta

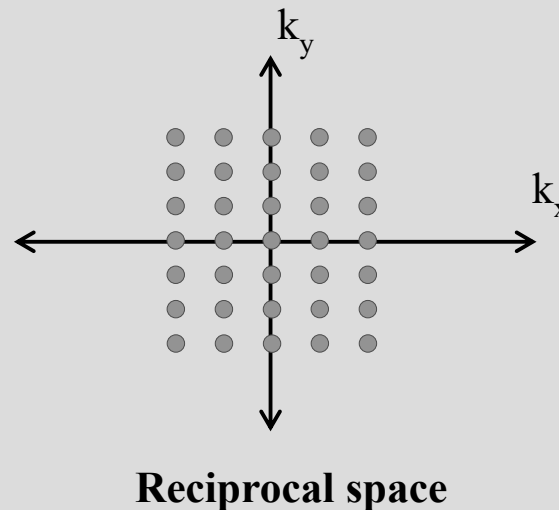
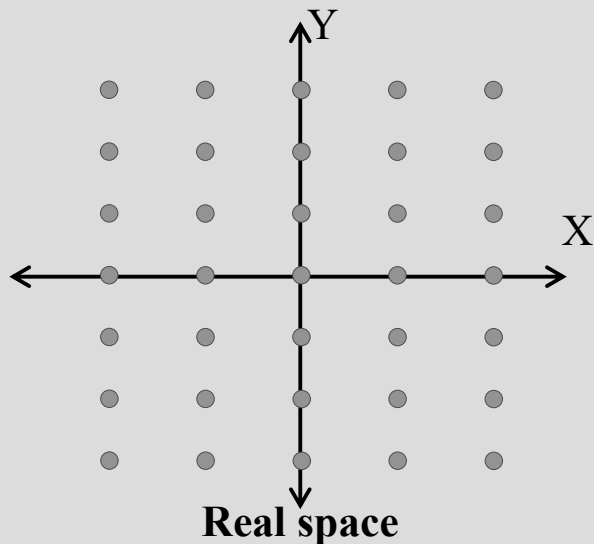
$\delta_{ij} = 1$ if $i=j$

$\delta_{ij} = 0$ if $i \neq j$

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)},$$

$$\mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)},$$

$$\mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}.$$



Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

2

Infinitely large basis set for a single wavefunction

Number $\mathbf{k}$ are restricted to only those inequivalent values which lie within a volume in the *reciprocal space* which is known as the *Brillouin Zone* (BZ).

$$\psi_{\mathbf{k}} = \psi_{|\mathbf{k}+\mathbf{G}|}$$

$$\psi_{\mathbf{k}+\mathbf{G}}(\mathbf{r}) = \sum_{\mathbf{G}} e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}} c_{\mathbf{k}+\mathbf{G}}$$

But $\mathbf{G}$ are infinite.

Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

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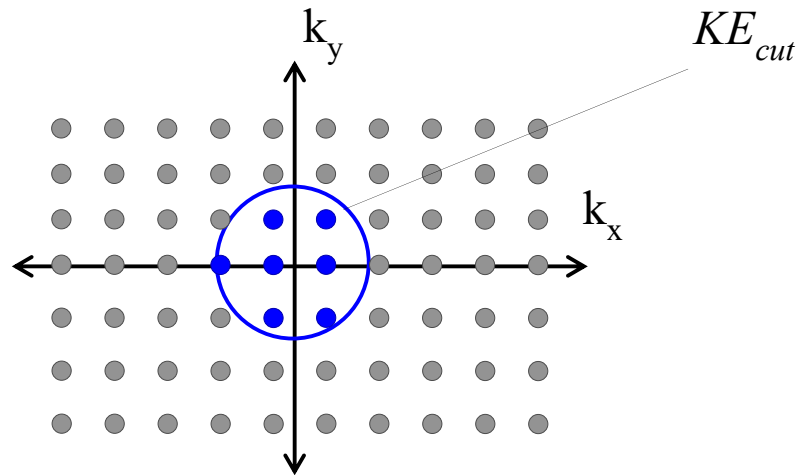
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But $\mathbf{G}$ are infinite.

Truncating condition:

$$|\mathbf{k}+\mathbf{G}|^2 \leq KE_{cut}$$



Supplementary Discussion

What is e-e exchange interaction?

- The electronic wavefunctions are antisymmetric nature.
- As a result, a spatial separation is produced between two electrons of same spin. This causes reduction in the Coulomb energy of the system.
- This reduction in energy is known as **exchange energy**.

Antisymmetric wavefunction:

For a system of two identical particles, a wavefunction (Ψ) is antisymmetric if:

$$\Psi(\mathbf{r1}, \mathbf{r2}) = -\Psi(\mathbf{r2}, \mathbf{r1})$$

where $\mathbf{r1}$ and $\mathbf{r2}$ represent the complete set of coordinates (including spin) for each particle.

What is e-e correlation interaction?

- The Coulomb energy can be reduced further if electrons with opposite spins are also spatially separated.
- In this case the Coulomb energy is reduced at the cost of increasing the kinetic energy of the electrons.
- The increase in kinetic energy is called **correlation energy**.

Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

3

Determining $v_{xc}[\rho]$

$$v_{\text{eff}} = v_s + v_c[\rho] + v_{xc}[\rho]$$

Local Density approximation (LDA):

Exchange-correlation energy of an inhomogeneous system can be approximated as sum of the exchange-correlation energy of a small regions which are locally homogeneous.

$$E_{xc}[\rho(\mathbf{r})] = \int \epsilon_{xc}[\rho(\mathbf{r})] \cdot \rho(\mathbf{r}) \cdot d\mathbf{r}$$

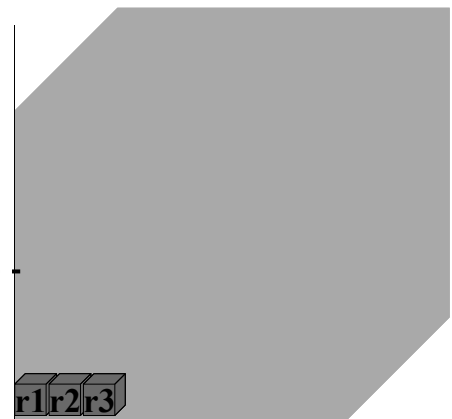
Works well for bulk system!

Generalized Gradient approximation (GGA):

Improvement over LDA by introducing semi-local term.

$$E_{xc}[\rho(\mathbf{r})] = \int \epsilon_{xc}[\rho(\mathbf{r}), \nabla \rho(\mathbf{r})] \cdot \rho(\mathbf{r}) \cdot d\mathbf{r}$$

Improves results generally but, especially for aperiodic systems!



Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

3

Determining $v_{xc}[\rho]$

$$v_{eff} = v_s + v_c[\rho] + v_{xc}[\rho]$$

Hatree: No exchange-correlation potential

Hatree-Fock: E_x is exact but no correlation

LDA: $E_{xc} \sim \rho(\mathbf{r})$

GGA: $E_{xc} \sim \rho(\mathbf{r}), \quad \rho(\mathbf{r})$

Meta-GGA: $E_{xc} \sim \rho(\mathbf{r}), \quad \rho(\mathbf{r}), \quad \rho^2(\mathbf{r})$

Hybrid-GGA: $E_{xc} \sim E_x[\text{Hatree}] + \text{correlation}$

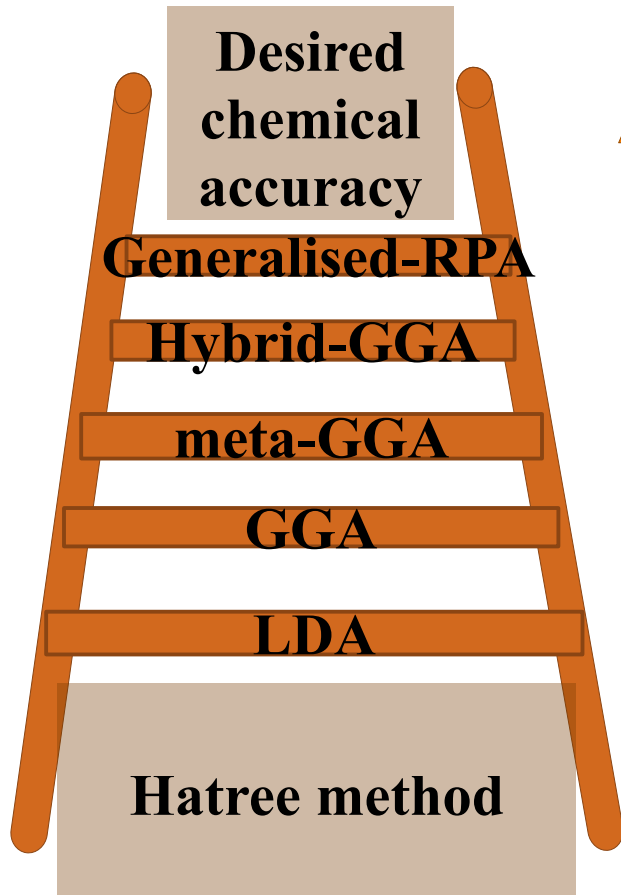
Generalised-RPA: $E_{xc} \sim \psi_i(\mathbf{r})$

❑ There is no one functional which works well for all systems.

❑ Some benchmark studies must be performed where results from more than one functionals are compared with available experimental values.

❑ Choosing right functional is non-trivial.

Jacob's Ladder



Computational Time

Understanding Density Functional Theory

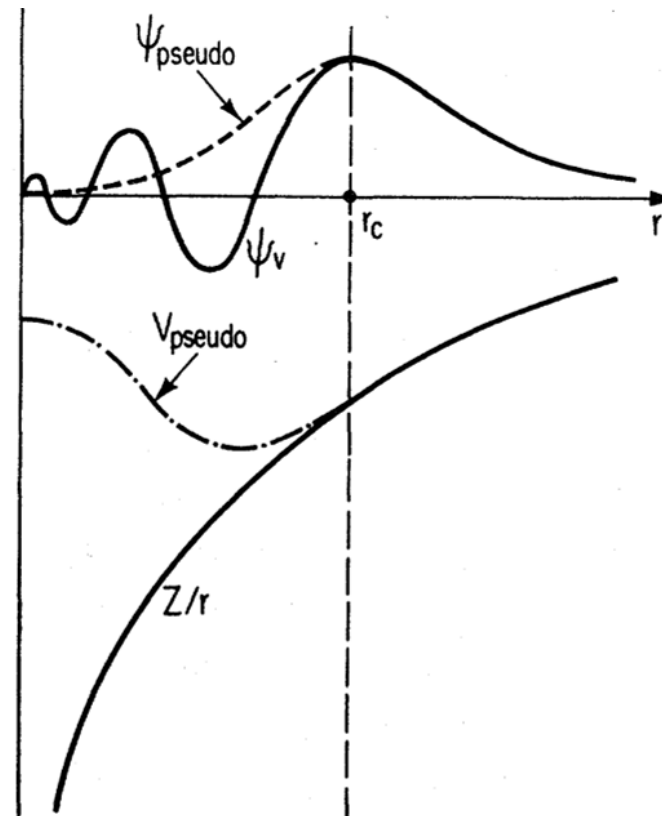
Formulation of DFT: Kohn-Sham Equations

4

Determining $v_s(r)$

$$v_{\text{eff}} = v_s + v_c[\rho] + v_{xc}[\rho]$$

- The electron wave functions near the nuclei shows rapid oscillation.
- Hence, large number of rapidly oscillating plane wave functions with very large E_{cut} is required.
- To reduce the computational cost, pseudopotentials (PPs) are used for accounting the ion-electron interaction in the core region.



Understanding Density Functional Theory

Formulation of DFT: Kohn-Sham Equations

4

Determining $v_s(r)$

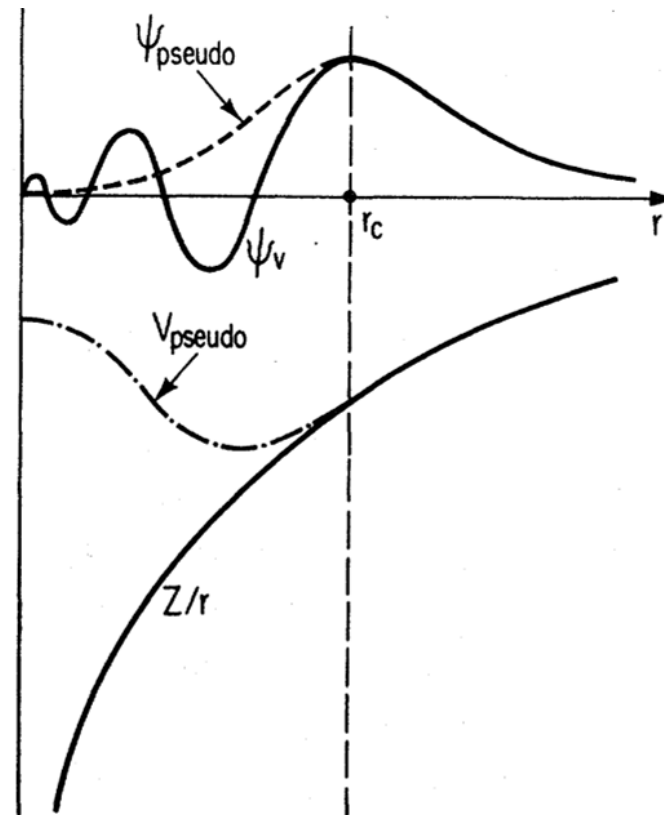
$$v_{\text{eff}} = v_s + v_c[\rho] + v_{xc}[\rho]$$

Norm-Conserving Pseudopotential

- $\int_0^{r_c} \psi_{\text{PS}}^* \cdot \psi_{\text{PS}} dr = \int_0^{r_c} \psi_{\text{AE}}^* \cdot \psi_{\text{AE}} dr$
- Charge density in core conserved
- Hard cut off radius and hence larger basis set

Ultrasoft Pseudopotential

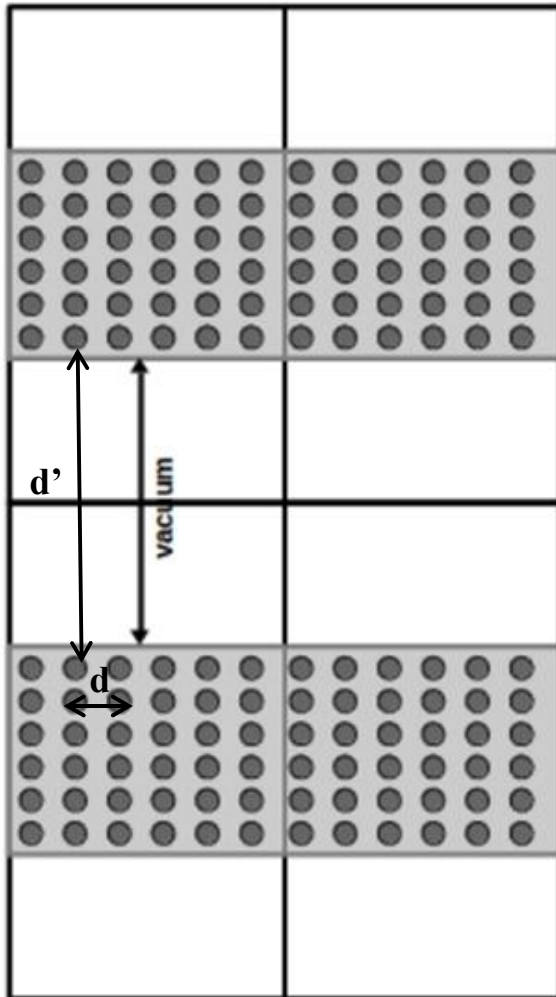
- Norm conserving condition is lifted
- Softer cut off radius
- Larger value for energy cutoff for charge density (ecutrho)



Understanding Density Functional Theory

Modeling non-periodic systems: Supercell construction

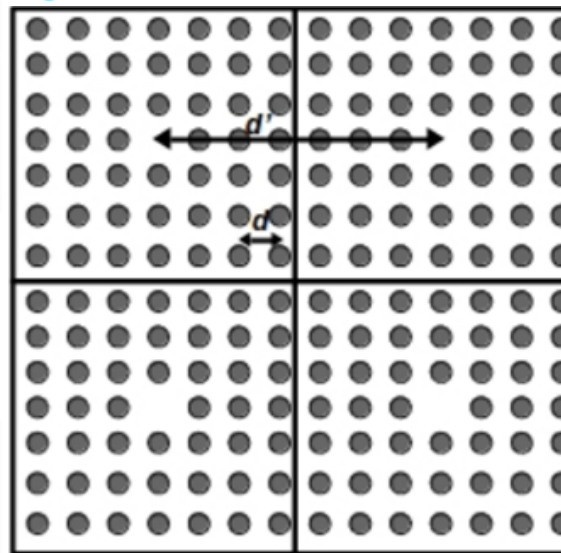
1 Surface



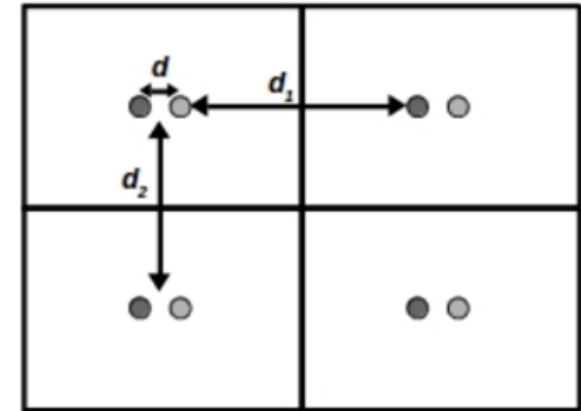
d : inter-atomic distance
 d' : lattice parameter

Condition for non-periodicity:
 $d \ll d'$

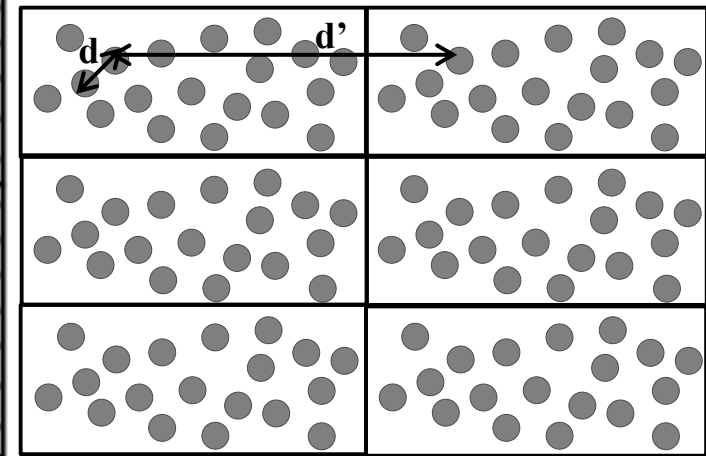
2 Point defect



3 Cluster/molecule



4 Amorphous solid



Input file for full geometry optimization: Diamond bulk

```
&control
  calculation = 'vc-relax',
  ! restart_mode='restart',
  prefix='lp',
  tstress = .true.
  tprnfor = .true.
  pseudo_dir = '.'
  outdir='.'
/
&system
 ibrav= 2,
nat= 2, ntyp= 1,
celldm(1)=6.7500000
ecutwfc =30
ecutrho =180,
/
&electrons
  conv_thr = 1.0d-8
  mixing_beta = 0.2
  mixing_mode='local-TF'
/
&IONS
/
&CELL
cell_dofree = 'ibrav'
/
ATOMIC_SPECIES
C 12.0110 C.pbe-rrkjus.UPF
ATOMIC_POSITIONS (crystal)
C 0.000000000 0.000000000 0.000000000
C 0.250000000 0.250000000 0.250000000
K_POINTS {automatic}
4 4 4 0 0 0
```

Variable cell relaxation

FCC lattice

Lattice parameter in Bohr unit

Full relaxation keeping the lattice same as defined

```
&CONTROL
  description of
  the calculation
/
&SYSTEM
  defining structural and
  electronic parameters
/
&ELECTRONS
  parameters for
  electronic cycle
/
&IONS
  parameters for
  ionic cycle
/
&CELL
  parameters for
  cell optimization
/
ATOMIC SPECIES
  name, atomic wt., PP file
ATOMIC_POSITIONS
  atomic coordinates
K_POINTS
  k-point mesh
CELL_PARAMETERS
  cell vectors
```

Structure of QE i/p file

Input file for full geometry optimization: Diamond bulk

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&control
  calculation = 'vc-relax'
  ! restart_mode='restart',
  prefix='lp',
  tstress = .true.
  tprnfor = .true.
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  outdir='.'
/
&system
 ibrav= 2,
nat= 2, ntyp= 1,
celldm(1)=6.7500000
ecutwfc =30
ecutrho =180,
/
&electrons
  conv_thr = 1.0d-8
  mixing_beta = 0.2
  mixing_mode='local-TF'
/
&IONS
/
&CELL
  cell_dofree = 'ibrav'
/
ATOMIC_SPECIES
C 12.0110 C.pbe-rrkjus.UPF
ATOMIC_POSITIONS (crystal)
C 0.000000000 0.000000000 0.000000000
C 0.250000000 0.250000000 0.250000000
K_POINTS {automatic}
4 4 4 0 0 0
```

*Variable
cell
relaxation*

FCC lattice

*Lattice parameter
in Bohr unit*

*Full relaxation
keeping the lattice
same as defined*

Use the input file for optimization calculation with executable 'pw.x'.

Quantities evaluated: Stress, forces on atoms and corresponding energy

Depending on the defined convergence thresholds the calculation either proceeds with updated cell parameters or terminates.

Input file for full geometry optimization: Diamond bulk

```
&control
  calculation = 'vc-relax'
!   restart_mode='restart',
  prefix='lp',
  tstress = .true.
  tprnfor = .true.
  pseudo_dir = '.'
  outdir='.'
/
&system
  ibrav= 2,
  nat= 2, ntyp= 1,
  celldm(1)=6.7500000
  ecutwfc =30
  ecutrho =180,
/
&electrons
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  mixing_beta = 0.2
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&CELL
  cell_dofree = 'ibrav'
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ATOMIC_SPECIES
C 12.0110 C.pbe-rrkjus.UPF
ATOMIC_POSITIONS (crystal)
C 0.000000000 0.000000000 0.000000000
C 0.250000000 0.250000000 0.250000000
K_POINTS {automatic}
4 4 4 0 0 0
```

Variable cell relaxation

FCC lattice

Lattice parameter in Bohr unit

Full relaxation keeping the lattice same as defined

Run the input file and
explore!!!